Assignment 2

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Preparation

```r
# packages
library("brms")
library("ggplot2")
library("tidyverse")

# data
load("data/income.RData")
```

data: `income.RData`

```r
head(income)
```

```
##    ls emotst inc_hh emotst_cat
## 1 3.8    6.0      7       high
## 2 1.0    3.0      6        low
## 3 5.0    4.0     10        low
## 4 4.2    3.5      7        low
## 5 4.4    6.0      2       high
## 6 3.8    6.5      7       high
```

The data set contains four variables

Variable	Description
`ls`	Life satisfaction (1-5)
`emotst`	Emotional stability (1-7)
`emotst_cat`	Emot. stability (high-low)
`inc_hh`	Household income (in 10.000)

Excercise 1

Fit the following Bayesian model to assess the effect of household income on life satisfaction.

```r
# Casewise exclusion of missing data
income <- na.exclude(income)

# This function can be used to figure out the structure of the
# priors for the sepcific model including names of the predictors
# default_prior(ls ~ inc_hh,
#               data = income)

bprior <- c(brms::prior(normal(3, 2), class = Intercept)
            , brms::prior(normal(0, 1), class = b, coef = inc_hh)
            , brms::prior(normal(0, 2.5), class = sigma))
```

```r
model.1 <- brm(ls ~ 1 + inc_hh
               , data = income
               , prior = bprior
               , silent = 2
               , refresh = 0)
```

```
## Trying to compile a simple C file
```

```
## Running /usr/lib/R/bin/R CMD SHLIB foo.c
## using C compiler: ‘gcc (Ubuntu 15.2.0-4ubuntu4) 15.2.0’
## gcc -std=gnu23 -I"/usr/share/R/include" -DNDEBUG   -I"/home/julia/R/x86_64-pc-linux-gnu-library/4.5/R
## In file included from /home/julia/R/x86_64-pc-linux-gnu-library/4.5/RcppEigen/include/Eigen/Core:19,
##                  from /home/julia/R/x86_64-pc-linux-gnu-library/4.5/RcppEigen/include/Eigen/Dense:1,
##                  from /home/julia/R/x86_64-pc-linux-gnu-library/4.5/StanHeaders/include/stan/math/pri
##                  from <command-line>:
## /home/julia/R/x86_64-pc-linux-gnu-library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10:
##   679 | #include <cmath>
##       |          ^~~~~~~
## compilation terminated.
## make: *** [/usr/lib/R/etc/Makeconf:202: foo.o] Error 1
```

a) Consider the priors. Do you think they are a good choice? Run a prior prediction and change the priors if necessary.

b) Fit the model with the adjusted priors. Ensure convergence and visualize the posterior distributions of all parameters. Interpret the results.

1c. Conduct a sensitivity analysis by fitting the model using three different priors.

1d. Run a posterior prediction. Visualize the results.